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Solid State Battery

Pilot line sample testing in progress; Expert call takeaways

Asia Autos & EV Battery

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Following a leading battery equipment maker's bullish comments on solid-state battery (SSB) orders at its 2Q25 result call, the Chinese solid state battery thematic stocks have rallied 50-110% from 1st Aug to a recent peak (CSI300 +17%). Companies participating in MIIT's Rmb6bn SSB project have recently submitted their samples from pilot lines for testing. However, all testing results may not be satisfactory, according to an industry expert working on one of the projects. While the stock market has seen a few rounds of enthusiasm around SSB in the past two years, we keep a more conservative view and do not expect commercial-scale production in the near future (see *Solid-state vs Sodium-ion Battery: Could CATL's Robin Zeng be right again?* published Jan-2025). In this report, we include expert call takeaways and the latest SSB progress of Chinese battery, material and equipment players. *CATL remains our top pick within China's battery value chain. We believe CATL will maintain its technology leadership as next-generation battery innovations emerge, whether in solid-state batteries or alternative advanced technologies (e.g. anode-free).*

- **ASSB pilot line samples being tested now.** Battery makers participating in MIIT's ASSB subsidy program have recently submitted samples from pilot production lines for testing. This was delayed by ~1 month due to production hiccups at select players.
- **Sample testing result may not be satisfactory.** While final results from sample testing are expected in Feb-Mar 2026, the expert speaker has the understanding that current pilot line samples still face challenges and may not be able to pass all the safety tests.
- **Safety features challenged for ASSB – Is solid-state battery truly needed for EVs?** The expert highlighted that sulfide electrolytes in ASSBs decompose at ~200°C and are prone to ignition due to their sulfur and phosphorus content, exhibiting behavior similar to liquid electrolytes. Additionally, the requirement for high-pressure operation poses significant challenges for integration into passenger vehicles. While ASSBs were originally developed to offer superior safety and energy density, the safety advantage is now in question, as there are no meaningful differences compared to liquid lithium batteries. Furthermore, low production yields and high costs may not justify the level of energy density improvement (14–25% over liquid lithium batteries using the same cathode system).
- **Semi-solid-state batteries in EVs are more of a marketing strategy than a practical breakthrough.** Many companies are shifting resources toward semi-solid-state solutions, which face lower technical barriers. In

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terms of energy density, semi-SSBs do not offer a meaningful improvement over liquid batteries using the same cathode materials. While there is a slight enhancement in single-cell safety, system-level safety gains remain limited. Consequently, EV OEMs often promote semi-solid-state batteries to advertise their models rather than to deliver significant advancements in safety or energy density.

- **Government support for SSB research may decline** following the initial Rmb6 billion project, as numerous challenges and difficulties have emerged over the past year.
- While CATL has been less vocal about its ASSB development compared to other battery makers who have attracted more media attention, the expert believes **CATL remains the most credible domestic contender in ASSB**. This confidence is supported by CATL's team of over 2,000 dedicated researchers and more than 100 parallel sub-teams focused on addressing various failure modes.
- **National standards on SSB emerging.** On 22-May 2025, the China Automotive Engineering Society (CAES) published the group standard titled *"All-Solid-State Battery Determination Method"*. The standard aims to provide clarity to longstanding confusion in the industry surrounding the definitions of "all-solid," "semi-solid," and "hybrid" battery types. The expert commented that the China Automotive Research Center is simultaneously preparing **national standards** likely to classify batteries into (i) All-Solid-State, (ii) Solid-Liquid Hybrid, and (iii) Liquid-State. National standards are expected to be more stringent and exclude "semi-solid" as a formal category.
- **CATL's condensed battery and anode-free technology are poised for commercialization in the near future.** While ASSB commercialization remains a distant goal, CATL is demonstrating that it is not the only path to achieving high energy density. In Apr-2023, CATL launched its condensed battery—akin to a "semi-SSB"—with an impressive energy density of 500 wh/kg. Further in Apr-25, the company unveiled its self-forming anode technology, which further increases energy density by up to 60% volumetric and 50% gravimetric. Although challenges such as shorter cycle life and lower charge rates persist, CATL's hybrid battery solutions are paving the way for practical adoption, with commercial production expected to begin in 2026.

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SSB Expert Call Takeaways

We include key takeaways from our recent expert call with a researcher leading sulfide-based SSB research at a state-owned institute in this note.

- **ASSB pilot line samples being tested now:** The battery makers participating in MIIT's ASSB subsidy program have recently submitted samples from pilot lines for testing. Per requirement from the MIIT program, each participating company must provide ASSB battery cell sample randomly picked from the pilot production line. These samples should be similar to the product that could be installed in EVs in 2027 (e.g. large battery cell at 60Ah). Cell testing was initially scheduled to begin in Sep-Oct 2025 but has been postponed to November, due to sample readiness issues at Geely and FAW. All participating companies (CATL, BYD, Geely and FAW for sulfide-based ASSB) have now submitted their samples to China Automotive Research Center on 7-Nov.
- **Testing results expected in 1Q26.** The review committee has not disclosed a formal timeline, but final evaluation results are expected around Feb-Mar 2026, with no official public announcement anticipated.

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- **What are the criteria for ASSB performance?** MIIT's SSB program has set out comprehensive metrics for the participating companies to meet. Below are the key metrics for sulfide-based ASSB:
  - **Energy density:** 400wh/kg gravimetric energy density and 1,000wh/L volumetric energy density.
  - **Safety tests:** Thermal safety assessments such as withstanding temperatures above 200°C in a hot-box chamber, and nail-penetration testing.
  - **Cycle times:** Over 1,000 charge–discharge cycles with at least 80% capacity retention. Cycling tests must be conducted under 10–20MPa external pressure to maintain solid–solid interfacial contact.
- **Technology routes and player participation**: There are six companies participating in the MIIT SSB program, including:
  - **Sulfide-based ASSB:** CATL, BYD, Geely, and FAW
  - **Polymer-based ASSB:** SAIC Qingtao and Welion
- **Sample testing result may not be satisfactory, according to the expert speaker's understanding**.
  - **ASSBs may not be entirely safe, which undermines their original purpose.** The expert speaker underscored that sulfide electrolytes decompose at ~200°C and inevitably ignite due to sulfur and phosphorus content, similar to liquid electrolyte behavior. Both nail penetration (induced short-circuit heating) and the 200°C hot-box test consistently trigger thermal runaway in sulfide systems.
  - **ASSBs require operation under high pressure, presenting significant challenges for their application in passenger vehicles.** Battery makers must also provide dedicated pressure fixtures for testing, and the expert emphasized that vehicle integration of such pressure requirements is extremely challenging—potentially feasible for heavy-duty trucks (HDTs) but largely impractical for passenger vehicles (PVs) due to space and system-design constraints.
  - **The balance between energy density and safety.** Current samples of sulfide-based ASSB can meet the energy density requirement but do not meet the safety requirements. The expert believes the 400wh/kg and 1,000wh/L energy density requirements are only attainable with the sulfide route. If the energy density requirement were relaxed, halide-based ASSB could meet the thermal-safety criteria and perform better than sulfide-based ASSB on safety. The halide route has not reached sample testing stage, as current halide materials still suffer from poor air stability and inadequate ionic conductivity, representing clear technical bottlenecks that prevent any halide-based products from being submitted.
- **Cathode and anode materials:** Delivering 400wh/kg is technically feasible with 9-series high-nickel ternary cathodes and ~50% silicon-carbon composite anodes. However, the expert noted that such high-nickel loading amplifies thermal safety risks, and achieving consistent production yield has proven challenging even for CATL and BYD. He emphasized that energy density improvements are primarily driven by cathode materials, not the electrolyte format. Current 9-series NCM liquid battery cells reach energy density at ~320–350wh/kg.

- **Government support for ASSB research may decline** following the initial Rmb6 billion project, as numerous challenges and difficulties have emerged over the past year.
- **ASSB commercialization on EVs still has a long way to go.** The expert views mass-production viability as distant. Material costs remain >10x higher than liquid-based systems, and yield/consistency problems are acute even for the leading players. Given intensifying EV price competition in China, he views mass deployment in mainstream EVs as highly unlikely. Limited volumes may appear in premium EVs by 2027, but he considers widespread adoption in 2030 very difficult.
- **Semi-solid-state batteries in EVs are more of a marketing strategy than a practical breakthrough.** Many companies are shifting resources toward semi-solid-state solutions, which face lower technical barriers—often utilizing ceramic-coated separators to enhance local thermal stability. In terms of energy density, semi-SSBs do not offer a meaningful improvement over liquid batteries using the same cathode materials. While there is a slight enhancement in single-cell safety, system-level safety gains remain limited. Consequently, EV OEMs often promote semi-solid-state batteries to advertise their models rather than to deliver significant advancements in safety or energy density.
- **Anode-free technology progress remains limited domestically:** While Korean players (e.g., SDI) demonstrated early progress in 2022, domestic players struggle with lithium deposition and rapid shorting. Most lab-level demonstrations depend on imported carbon powders. The expert noted that while CATL and several suppliers claim technical readiness, stable manufacturability remains unresolved.
- **Assessment of domestic players beyond CATL/BYD:** Material-side innovation is currently the most tangible area of domestic progress. Several tier-2 players have secured other national projects, but the expert viewed their execution as weak—citing poor build quality in an 80Ah ASSB cell that was debuted on the market recently and slow, under-resourced line commissioning at other tier-2 battery makers. The expert believes neither will surpass CATL or BYD.
- **CATL remains the most credible domestic contender in ASSB**, backed by >2,000 dedicated researchers and over 100 parallel sub-teams tackling failure modes.
- **Evolving standardization landscape:** On 22-May 2025, the China Automotive Engineering Society (CAES) published the group standard titled *"All-Solid-State Battery Determination Method"* (全固态电池判定方法). The standard aims to provide clarity to longstanding confusion in the industry surrounding the definitions of "all-solid," "semi-solid," and "hybrid" battery types. The standard establishes a weight-loss (mass loss) test method to quantify residual liquids: after a visual inspection for absence of liquid leakage, the sample is vacuum-dried (for instance at 120°C for some protocols) and if the mass loss is below 1%, the cell can be deemed "all-solid." Expert commented that China Automotive Research Center is simultaneously preparing **national standards** likely to classify batteries into (i) All-Solid-State, (ii) Solid-Liquid Hybrid, and (iii) Liquid-State. National standards are expected to be more stringent and exclude "semi-solid" as a formal category.

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CATL's Anode-Free Technology

CATL's announcement

CATL officially unveiled its anode-free battery technology, also known as self-forming anode technology, on April 21, 2025, during its inaugural Super Tech Day. This innovation was introduced as one of three major battery technologies - the Freevoy Dual-Power Battery (integrates self-forming anode technology), Naxtra - the world's first mass produced sodium-ion battery, and the second-generation Shenxing Superfast Charging Battery.

Although the anode-free concept has long existed, particularly in lithium-metal and sodium-ion battery research, widespread commercialization had been hindered by challenges such as dendrite growth, morphology instability, and interfacial degradation.

CATL's implementation is based on self-forming anode architecture, where no traditional anode active material (e.g., graphite or hard carbon) is used during initial assembly. Instead, a bare metal current collector (typically copper or aluminum foil) serves as the "nominal anode." During the first charge, lithium (or sodium) ions migrate from the cathode and electrochemically deposit onto the current collector, forming an active metal anode in situ.

CATL's self-forming anode technology is a platform solution that can be applied to various metal batteries, including sodium, lithium, zinc, and potassium systems. This technology replaces traditional graphite anodes entirely, leading to a significant increase in battery energy density. The "self-forming anode technology" represents a disruptive breakthrough at the atomic level, as the volumetric energy density of the battery can increase by 60%, and the gravimetric energy density by 50%. This also means that more power can be engineered into the same battery pack space, supporting longer range. This technology can be flexibly paired with various material systems, and when combined with NCM systems, the energy density can be increased to over 1000Wh/L. The technology also enables the creation of a "battery within a battery" design, like the Freevoy Dual Power Battery, which combines lithium-ion and sodium-ion cells for optimized performance.

What is anode-free technology?

Anode-free technology refers to a next-generation lithium metal battery architecture in which no lithium-based anode material is present during initial cell assembly. Instead, a bare copper foil is used as the current collector. During the first charging cycle, lithium ions deintercalated from the cathode and electrochemically deposit onto the copper surface to form a functional lithium metal anode. As a result, this type of battery is also known as a lithium-free or anode-free cell.

CATL's self-forming anode technology – what was published in the science journal *Nature Nanotechnology?*

On 28-May 2025, CATL's research team published a paper in the top-tier journal *Nature Nanotechnology* titled: *"Application-driven design of non-aqueous electrolyte solutions through quantification of interfacial reactions in lithium metal batteries"*. The paper elaborates on the optimization of electrolyte system and details the cycling performance of anode-free batteries.

The research shows that both lithium-metal and anode-free batteries use a new electrolyte system, named **LiFSI–1.2DME–3TTE**, which is vastly different from the conventional **LiPF6–EC/EMC** system.

- **DME (1,2-Dimethoxyethane)**: A common ether solvent that significantly reduces viscosity and improves fluidity.
- **TTE (1,1,2,2-Tetrafluoroethyl-2,2,3,3-tetrafluoropropyl ether)**: A fluorinated ether solvent that mitigates corrosion of the aluminum foil cathode current collector — a problem mainly caused by LiFSI.

The key optimization lies in the choice of lithium salt solvent systems, aiming to:

- Improve reduction resistance
- Enable formation of a more stable SEI (solid electrolyte interphase) on the lithium surface
- Enhance the mechanical strength and flexibility of the SEI to accommodate volume changes during charge-discharge cycles, thus prolonging battery life

Challenges of self-forming anode technology
Based on the material system mentioned, a 50Ah pouch cell could reach 400Wh/kg, which is more than twice the energy density of LFP batteries, or 1.5 times higher than typical NMC batteries, offering clear advantages in extending EV driving range.

Despite these improvements, it is likely that technical challenges remain. For instance, the cycle life of the 0μm lithium metal (anode-free) battery can be only around 120 cycles, significantly lower than the 500+ cycles of conventional 20μm lithium metal batteries. Additionally, the charging rate is only 0.2C, much lower than the ~2-3C+ typical of lithium-ion batteries.

How can we solve the challenges of short cycle life? On one hand, further improvements are needed; on the other, a **hybrid battery system** is being considered, where the self-forming anode cell acts as a range extender and most of the cycling is handled by the main battery — a compromise solution at this stage.

CATL's application of self-forming anode technology
The Freevoy Dual-Power is a breakthrough product that deeply integrates CATL's dual-power architecture and self-forming anode technology. The dual-power architecture means that the battery pack has two powerful independent energy zones, which enables five dual functions: dual high-voltage, dual low-voltage, dual structure, dual thermal management, and dual thermal runaway safety protection, ensuring the continuity, stability and safety of power output. This dual-power design and innovative integration of software will provide a more stable and reliable power supply for vehicles in the upcoming L3 and L4 autonomous driving era.

The "dual-electric range extension" technology pioneered by the Freevoy Dual-Power Battery can intuitively regulate the allocation strategy of two energy zones based on the vehicle's driving status and users' driving habits. The main energy zone of the dual-power battery can use different chemical systems of battery cells according to the users' driving habits and scenarios, meeting daily driving needs; The extended range energy zone can adopt high specific energy self-forming anode technology to provide greater capacity to meet users' long-distance travel needs.

Three dual-power solutions across different chemical systems were released:

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- **Sodium-LFP Dual-Power Battery:** It combines Naxtra with a LFP self-forming anode battery, fully utilizing the low-temperature performance of sodium-ion technology to provide users with an exceptional experience that excels in cold conditions while delivering extended range.
- **LFP-LFP Dual-Power Battery:** It pairs the second-generation Shenxing Superfast Charging Battery with the LFP self-forming anode battery. It easily achieves 1,000km of pure electric range in sedans with a 3-meter wheelbase, reducing the commuting cost per km to as low as Rmb0.1.
- **NCM-LFP/NCM-NCM Dual-Power Battery:** It integrates an NCM battery with an LFP self-forming anode battery, achieving a peak charging rate of 12C for the NCM battery in the main energy zone, providing over 1 megawatt of power. Even when SOC drops to 20%, it can still output over 600kw of power. The upgraded version of the product, consisting of an NCM battery and an NCM self-forming anode battery enables a capacity of over 180kwh in sedans with a 3-meter wheelbase, breaking through the 1,500km pure electric range barrier.

Global battery makers' progress of self- forming anode technology

We summarize global battery makers' progress of self-forming anode technology below. We note CATL's self-forming anode approach highlights the critical contribution of the electrolyte system, while other battery makers' self-forming anode technology emphasized on application on SSB and downplayed the role of liquid electrolytes.

Table 1: Global battery makers' progress on self-forming anode technology

Company	Disclosure Date	Key Technology	Key Differences vs. CATL
CATL	Apr 21, 2025, at Super Tech Day	Self-forming Li/Na metal anode on Cu/Al current collector compatible with multiple cathode systems (NCM, LFP, etc.) +60% volumetric & +50% gravimetric energy density	
Quantum Scape	May 1, 2025	Solid-state ceramic separator + factory anode-free design 15 min fast charging; launching with VW in 2025	Uses solid-state system relies on proprietary ceramic separator
Our Next Energy (ONE)	Sep 13, 2022	240 Ah prismatic anode-free cell 1007 wh/L energy density; dual-chemistry (LFP + anode-free) pack Mass production targeted in 2026	Higher energy density dual-chemistry pack reduces reliance on anode-free durability
Tesla (Dalhousie team)	Aug 2020	Electrolyte additive systems for anode-free Li-metal focus on improving CE and longevity	Still at lab/patent stage no public mass production path
Tailan	Nov 7, 2024 — Joint launch with Changan Auto	Solid-state/"no separator" cell launched under "4-3-2-1" roadmap (gradual elimination of separator, electrolyte, and finally anode)	Gradual anode elimination focuses on solid-state/hybrid roadmap
Jinyu	Jan 24, 2024	15Ah Li-metal anode-free cell semi-solid electrolyte; 420 wh/kg, 1100 wh/L already deployed in drones	First to commercially ship anode-free battery high energy density focused on niche UAV market

Source: Company data.

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Solid State Battery

Definition of solid-state battery updated

The definition of solid-state battery was updated in May 2025 based on the China Society of Automotive Engineers (CSAE)'s standard set in "*Judgment Method for All-Solid-State Batteries* (T/CSAE 434-2025)".

Judgment Method for All-Solid-State Batteries (T/CSAE 434-2025)

On May 22, 2025, the China Society of Automotive Engineers (CSAE) officially released the group standard *Judgment Method for All-Solid-State Batteries* (T/CSAE 434-2025), which for the first time clarifies the definition of solid-state batteries.

The standard aims to provide clarity to longstanding confusion in the industry surrounding the definitions of "all-solid," "semi-solid," and "hybrid" battery types. An "all-solid-state battery" is required to achieve ion transport exclusively through solid electrolytes, establishing a strict technical boundary with hybrid solid-liquid electrolyte batteries.

According to the standard, the tested sample must first undergo a qualitative inspection by visual observation of the breakage (no liquid seepage) to exclude obvious liquid residues, followed by a quantitative test where the weight loss rate after vacuum drying at 120°C for 6 hours is <1% (the weight loss rate refers to the ratio of the mass lost by the sample to its initial mass under specific conditions). The core research focus of this standard lies in proposing a test method for liquid substance content based on the weight loss rate. Supported by multiple rounds of verification tests, the test method of heating under vacuum conditions to measure the weight loss rate is determined as the basis for judging all-solid-state batteries.

The judgment method adopts a "two-step" detection process:

- Qualitative detection: No liquid seepage observed visually at the breakage.
- Quantitative detection: After vacuum drying at 120°C for 6 hours, the battery weight loss rate is <1% (weight loss rate = mass lost by sample / initial mass).

Test verification shows that this method has an error rate as low as 0.3%, covers mainstream technical routes such as sulfides, oxides, and polymers, and the test conditions (vacuum degree -0.095~-0.1 MPa) balance safety and universality.

J.P. Morgan comments: Under the new standard, cells containing less than 1% liquid electrolyte are still classified as ASSBs. We view this as a pragmatic compromise, reflecting ongoing challenges with interfacial stability.

National standard expected to be more stringent

China Automotive Research Center is simultaneously preparing national standards likely to classify batteries into (i) All-Solid-State, (ii) Solid-Liquid Hybrid, and (iii) Liquid-State. National standards are expected to be more stringent and exclude "semi-solid" as a formal category.

Chinese government's target on ASSB

MIIT's Rmb6bn ASSB subsidy program

China's Ministry of Industry and Information Technology (MIIT), together with other key ministries, launched a Rmb6bn special fund to support the development of solid-state batteries in May-2024. The initiative aims to help China maintain its leadership in next-generation battery technologies amid rising global competition, especially from Korean and Japanese players advancing in SSB.

The program selected six leading Chinese companies to receive funding for R&D, pilot production, and testing. The goal is to assess the technical feasibility of ASSB through standardized evaluations and de-risk the technology before full industrialization. There are six companies included:

- **Sulfide-based ASSB:** CATL, BYD, Geely, and FAW
- **Polymer-based ASSB:** SAIC Qingtao and Welion

Timeline of the program

The timeline includes cell testing in 2025, system level validation in 2026, and commercial deployment by 2027. The goal is for each company to achieve 1,000 vehicle installations with ASSB by 2027.

The Rmb6bn subsidy for solid-state batteries is seen as a trial effort to explore the feasibility of SSBs, particularly due to concerns about Korea and Japan potentially overtaking China in this technology. This effort is considered both a strategic push and a technology validation exercise. Depending on test results in 2025, government and private sector capital allocation may scale up or down accordingly.

Technical requirements

- **Energy density:** 400wh/kg gravimetric energy density and 1,000wh/L volumetric energy density.
- **Safety tests:** Thermal safety assessments such as withstanding temperatures above 200°C in a hot-box chamber, and nail-penetration testing.
- **Cycle times:** Over 1,000 charge–discharge cycles with at least 80% capacity retention. Cycling tests must be conducted under 10–20MPa external pressure to maintain solid–solid interfacial contact.

Current status & challenges:

- **Cell capacity:** Achievable and pilot cells are already produced, but **consistency remains poor**.
- **Safety tests (thermal and nail penetration tests) are very difficult to pass.** In order to achieve high energy density, SSBs still rely on high specific capacity cathode materials, which suffer from poor thermal stability.
- **Cycle life (1000+ cycles at 60Ah): Not yet achieved**; only small-format cells can meet this.

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Chinese government policy related to SSB

Table 2: Recent national-level policy directions related to SSB

Document Title	Date	Content Related to Solid-State Batteries
New Energy Vehicle Industry Development Plan (2021–2035)	2020.11	Accelerate the R&D and industrialization of power battery technologies such as solid-state batteries.
14th Five-Year Plan for New Energy Development Implementation Plan	2022.01	Accelerate the R&D and industrialization of power battery technologies such as solid-state batteries.
Medium- and Long-Term Science and Technology Development Plan for Carbon Peaking and Carbon Neutrality (2022–2030)	2022.06	Conduct research on key technologies such as solid-state batteries and all-solid-state battery systems.
Guiding Opinions on Strengthening the Construction of National Battery Innovation System	2023.01	Promote the development and application of solid-state battery technologies and strengthen research on solid-state battery system integration.
Major Research Program: Development of High-Safety, High-Efficiency Battery Materials Based on Next-Generation Components	2024.02	Develop high-safety and high-performance solid-state battery materials and key technologies.

Source: J.P. Morgan, National Natural Science Foundation of China (NSFC), State Council of the People's Republic of China, NDRC

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Battery makers' progress on SSB/ASSB

Table 3: Battery makers' SSB/ASSB progress

	Solid electrolyte	Cell level E/D (wh/kg)	Battery Content	Cycle times	Small scale production time	Mass production timeline	Anode	Electrolyte % in liquid form
China								
CATL	Sulfide	500	20Ah (current) 60Ah (2027)	1000	2027	n/a	Li-Metal	0%
CATL – condensed battery	Jelly-type electrolyte	500	n/a	n/a	2025 (eVTOL) 2027 (civil aviation)	n/a	Silicone	0% (2027)
BYD	Sulfide	400	60Ah	n/a	2027	2030-32	Silicone	0%
EVE	Polymer combined with Halide	300 (current) 400 (2025)	10Ah (current)	n/a	2027	n/a	Silicone	n/a
Gotion	Sulfide	350	70Ah	3000	2027	2030	Silicone	0%
Gotion	Polymer	300	70Ah	3000	2026-27	2028	Silicone	5%
CALB	Sulfide	430	50Ah	n/a	2027	2028	n/a	0%
Great Power	Oxide	300	20Ah	600	2025	2026	Silicone	0%
Sunwoda	Polymer	500 (current) 700 (2027)	60Ah (2026)	1000 (2026)	2026	n/a	n/a	0%
Farasis	Sulfide	400 (current) 500 (next gen)	60Ah	4000	2027	2030	Silicone	0%
Svolt	Sulfide	380	20Ah	n/a	2027	2030	Silicone	5-10% (current) 0% (2030)
Lishen 力神	Sulfide	375	20Ah	n/a	2027	n/a	Silicone	0%
Yinpai 因湃	Sulfide	400	30Ah	4000	2026	n/a	Silicone	0%
Yaoning 耀宁	Sulfide	400	20Ah	n/a	2027	n/a	Silicone	n/a
Welion 卫蓝	Oxide&Polymer	400 (2027)	10Ah (current)	n/a	2027	n/a	Li-Metal	<10% (current) 0% (2027)
Qingtao 清陶	Polymer Halide	406 (current) 500 (2026)	n/a	600	2026	2027	Silicone	5-15% (Gen 1) <5% (current) 0% (2027)
Inx Energy 欣界	Sulfide	450-550	1-55Ah	800-1000	2023	2025	Li-Metal	5%
Prologium 辉能	Oxide	321 (Gen1) 359 (Gen 2) 400+ (2025)	106Ah (Gen 1) 124Ah (Gen 2)	1000+	2028-29	n/a	Silicone Li-Metal (2025)	<10%
Talent 太蓝	Oxide&Polymer	400-500	n/a	n/a	2027	n/a	Silicone/ No anode	n/a
Jinyu 金羽	Sulfide	400	70Ah	n/a	2026-2027	2030	No anode	n/a (Quasi-solid battery)
GAC Aion	Sulfide	400	60Ah	n/a	2026	2030	Silicone	0%
Ex-China								
Toyota	Sulfide	350	30Ah	n/a	2027-2028	2030	n/a	0%
Honda	Sulfide	n/a	n/a	n/a	Est 2027	2030	n/a	0%
Hyundai	Sulfide	n/a	n/a	n/a	2025 onwards	2030	n/a	0%
Quantum Scape	Oxide	400	5Ah	1000	3Q24	2026	Li-Metal	0%
Solid Power	Sulfide	320 (2020) 430 (next gen) 560 (target)	2Ah (2020)	400	2026	2030	n/a	0%
BasqueVolt	Polymer & oxide	450	20Ah 80Ah (next gen)	n/a	n/a	2027	n/a	0%
SDI	Sulfide	450	20Ah (current) 90Ah (next gen)	1000	2027	2027-30	Li-Metal	0%
SKI	Sulfide	>400	n/a	n/a	2028-30	2030	Li-Metal	0%
LGES	Polymer & Sulfide	>400	n/a	n/a	2028-30	2030	Li-Metal	0%
Hitachi Zosen Corporation 日立造船	Sulfide	320	5Ah	n/a	n/a	2030	n/a	0%

Source: Company data, J.P. Morgan.

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Material and equipment makers on SSB/ASSB

Table 4: Supply chain investments in SSB

Equipment	SSB related progress and investments
Leading Chinese player	• Delivered SSB dry coating equipment to a **Korean client** in Nov-2024 • Based on breakthroughs throughout the whole process of solid-state battery manufacturing, company can provide customers with key equipment for dry mixing, dry film composite and solid-state stacking. • Delivered **the world's first automotive-grade all-solid-state battery turnkey solution**, demonstrating competitive advantages in this cutting-edge technology. • Company has been progressively delivering standalone key equipment for solid-state battery production to multiple customers around the world.
Hymson (688559 CH)	• Hymson has equity investment in **Inx Energy** and supply equipment to Inx Energy "Falcon" SSB. Signed 2Gwh SSB equipment orders with Inx Energy with total amount at Rmb400mn. Hymson supply whole production line SSB equipment to Inx Energy. • Won SSB equipment order from **Welion**. Target at small scale production in 2027 and large scale production 2028-30.
Lyric (688499 CH)	• Started R&D in SSB equipment since 2020 • Won SSB equipment orders from **Qingtao**. Delivery of SSB equipment in Jul-2022 and the production line has been put into production in mid-2023. Supply back-end equipment including formation and grading as well as laser welding equipment to Qingtao. • Delivered semi-SSB equipment to **Gotion's** semi-SSB trial production line in 2022 • Won Mwh level SSB trial production line order from **GAC Aion** Yinpai (total amount Rmb29mn). Target to deliver at early-2025. • Won one production line SSB equipment order from a domestic leading SSB player (**sulfide technology** route) in Nov-2024
Yinghe (300457 CH)	• Focus on SSB coating equipment & mastered solid-state battery wet coating technology, dry mixing technology, dry film forming technology, etc. • Company has delivered wet coating and dry mixing equipment to domestic customers recently. • Company has also launched the third-generation dry mixing + dry film-forming integrated product this year.
Naknor 纳科诺尔 (832522 CH)	• Company launched dry electrode forming and laminating integrated machine. • Company signed Rmb3bn letter of intent on SSB battery equipment with one domestic leading solid state battery maker in Jul-2024. • Company started collaboration with **Qingtao** in 2019 and supply SSB equipment to Qingtao.
Hangke (688006 CH)	• Hangke cooperates with Gotion on Gotion's 0.2Gwh trial production line for ASSB • Company has developed high pressure back-end formation and grading equipment
Putailai (603659 CH)	• Started R&D in SSB equipment since 2020 • Company has developed dry coating equipment • Company has sold dry coating equipment to CATL/BYD/SDI/SKI and Toyota • The first dry coating equipment was sold to Tesla • Company participated in MIIT's ASSB program, supplying ASSB equipment to FAW & Lishen • Company has won 100 ASSB orders, cumulative amount at Rmb200mn
Separator or solid electrolyte	
Yunnan Energy (002812 CH)	• Yunnan Energy has begun small scale production of high purity sulfide electrolytes (LPSC) through its Hunan Enjie Frontier subsidiary, now operating a pilot line capable of ton level annual output, with both sulfide and full solid electrolyte membrane samples under testing. • Company has a subsidiary working on ASSB separator films and have sent some samples to **Nissan**. • Yunnan Energy also has a JV with **Beijing Welion** on semi-solid-state battery and have product for Nio. • Company secured a long term contract with **Welion New Energy** to provide 300mn sqm of semi-solid state battery separators and at least 100 tons of solid state electrolyte materials from 2025 to 2030. Complementing that, a framework procurement agreement with Peking Weihui New Energy commits to delivering at least 0.3 billion m² of solid state battery electrolyte separators and electrolytes by 2030.
Senior (300568 CH)	• Company has equity investment in Xinyuanbang (新源邦). Xinyuanbang has a production capacity of hundreds of tons for **oxide electrolytes** and has achieved shipments at the ten-ton level. **Sulfide electrolytes** are expected to begin ton-scale shipments in 2025. Xinyunbang also supply sulfide electrolyte to BYD at small scale. • Company has signed an agreement with Dacao Chemical (大曹化学) to provide polymers for processing semi-solid battery separators. • For rigid frameworks and solid-state electrolyte membranes, the company has entered into a strategic partnership with CAS DeepBlue Huize to jointly develop market-ready **solid-state electrolyte membrane systems and high-performance all-solid-state batteries**, leveraging Senior's advanced rigid framework membrane technology. Recently, the company also signed a strategic cooperation agreement with Ruigu New Materials, which launched **China's first mass production line for sulfide electrolytes**, to jointly develop **high-performance sulfide electrolyte membranes** and related products, and accelerate commercialization. • At the 2025 Shenzhen International Battery Technology Exhibition, the company unveiled **polymer electrolyte separators and rigid framework products**, featuring industry-leading properties such as porosity close to 80%, micron-level pore size, high mechanical strength, and low thermal shrinkage. These products are suitable for next-generation semi-solid and all-solid-state batteries.
Qingtao (private)	• Company has successfully developed oxide solid-state electrolyte, company has 1300tons annual capacity of solid electrolyte
Easpring (300073 CH)	• Company has R&D in oxide and sulfide solid electrolyte • **ProLogium, Qingtao, Welion** and **Gangfeng** are solid electrolyte customers
Tianci (002709 CH)	• Company has solid electrolyte patent in sulfide solid electrolyte
Capchem (300037 CH)	• Capchem explores new solid-state electrolyte materials and preparation processes. For example, in polymer solid-state electrolytes, it studies the complexation of polymer materials such as polyethylene oxide (PEO) with lithium salts, and adds inert fillers to improve performance
Sanxiang Advanced	• Company conducts synthesis tests of zirconium-containing composite oxides (锆复合氧化物) and built a small-scale pilot line for zirconium-

Materials 三祥新材 (603663 CH)	based chloride materials (锆基氯化物材料) • The zirconium-based chloride materials have been preliminarily verified by customers and have been supplied to downstream solid-state battery makers in small batches • Customers include **CATL, Qingtao** etc.
Jinlongyu 金龙羽 (002882 CH)	• Company adopts oxide technical route and has completed the manufacture verification of 10kg class LATP solid-state electrolyte powder and slurry • Company has annual capacity of 10tons class LATP solid-state electrolyte • Solid-state electrolyte has been put into trial production
Ruitai New Materials 瑞泰新材 (301238 CH)	• LiTFSI produced by Ruitai is an important solid-state electrolyte material and can be used directly as the solid electrolyte
Ganfeng (002460 CH)	• Company mainly focuses on two technical routes: oxide and sulfide. The oxide solid electrolyte materials include two series of products with NASICON structure and Garnet structure. Four types of sulfide solid electrolyte products have also been developed. • Company has 200tons annual capacity of solid electrolyte.
Ronbay (688005 CH)	• Company successfully developed sulfide solid-state electrolyte and plan for trial production in 2025.
Tianmu Pilot 天目先导 (private)	• Company has 3000tons annual capacity of solid electrolyte, mainly used to supply **Welion**.
Blue Solid New Energy 蓝固新能源 (private)	• Company has successfully developed oxide and sulfide solid electrolyte • Company has designed annual capacity of 1) 55k tons in-situ solid-state electrolyte (原位固态电解质), 2) 7k tons of solid-state electrolyte powder, and 3) 10k tons of solid-state electrolyte slurry. Company's solid state electrolyte capacity is mainly used to supply **Welion**.
Xtc New Energy Materials (688778 CH) 厦钨新能	• Company has been actively expanding into the solid-state battery materials sector, focusing on the integration and compatibility of cathode materials with solid electrolytes. XTC has achieved initial small-scale production in halide and sulfide-based solid-state battery systems. • Company has entered into a framework cooperation agreement with Sunwoda's EV battery subsidiary to jointly develop and commercialize solid-state cathode materials, solid electrolytes, and electrolyte films, aiming to improve safety, ionic/electronic conductivity, and lithium dendrite suppression.
Solid Power (SLDP US)	• Focus on R&D and manufacture of sulfide solid electrolyte
NEI Corporation (private)	• NEI Corporation successfully developed sulfide-based, oxide-based, polymer-based and halide-based solid-state electrolyte
Ampcera (private)	• Actively developing, manufacturing and commercializing solid electrolytes globally with more than 200 paying customers. Ampcera's technology also includes cutting edge SSB technology backed by the U.S. DOE and in partnerships with both GM and Ford.
Anode (silicon anode or lithium metal anode)	
Ganfeng (002460 CH)	• Company's lithium metal anode product has entered **Qingtao** and **Tailan**'s (太蓝能源) supply chain. • Production bases for **lithium metal** have been established in Yichun and Fengxin, Jiangxi Province, with an existing production capacity of 2,150 tons. Additionally, part of the 1000tons lithium metal anode capacity in Qinghai has been put into trial production.
BTR (835185 CH)	• Focuses on the layout sulfide solid-state battery materials and **lithium metal composite anodes**
Putailai (603659 CH)	• Company has R&D in lithium metal anodes • Company has 120tons silicon carbon anode capacity in 2024 and expects to expand to 240tons – 960tons annual capacity in 2025.
Shanshan (600884 CH)	• Company has designed 40k tons integrated silicon-based anode annual capacity in Ningbo. The first phase production capacity is scheduled to be put into trial production by end of 2024.
Shangtai (001301 CH)	• Company forms a dedicated technical team for R&D of silicon-carbon anodes • Company has built a silicon-carbon anode pilot production line
Dowstone 道氏技术 (300409 CH)	• Company has sent silicon-carbon anode samples to **Tailan (**太蓝能源) and **Mengul**i (盟古利).
Shanghai Xiba 上海洗霸 (603200 CH)	• Company announced in June 2024 that it plans to invest Rmb380mn in building silicon-carbon anode capacity.

Source: Company data, J.P. Morgan.

SSB demand forecast by GGII

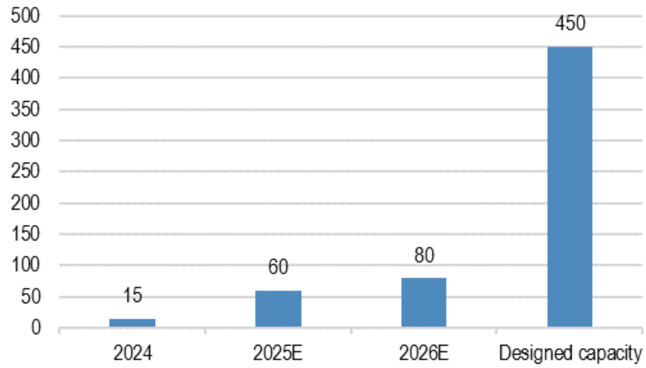
Capacity: GGII expects 80Gwh capacity in 2026, up from 15Gwh in 2024

The industry has >15Gwh of effective capacity of solid-state battery in 2024 and Gaogong Industry Research Co., Ltd (GGII) expects the total production capacity of

SSB (including ASSB and semi-SSB) to reach 80Gwh in 2026E (vs planned capacity at >450Gwh).

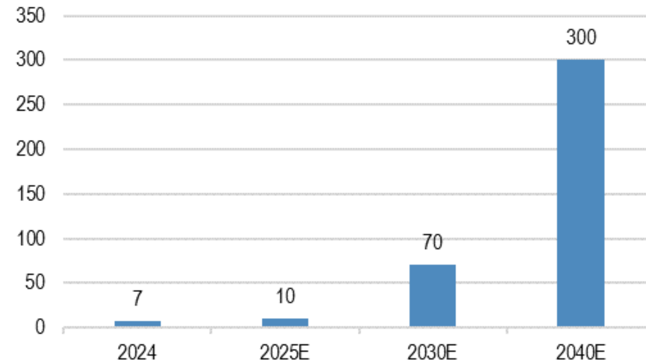
In terms of demand, GGII estimates 70Gwh SSB demand in 2030E including ASSB and semi-solid state battery, up from ~10Gwh this year. Another third-party consulting firm, ICCSino, estimates 80Gwh in 2030E including ASSB and semi-solid state battery, up from ~5Gwh this year.

Figure 1: Solid-state battery capacity (Gwh)



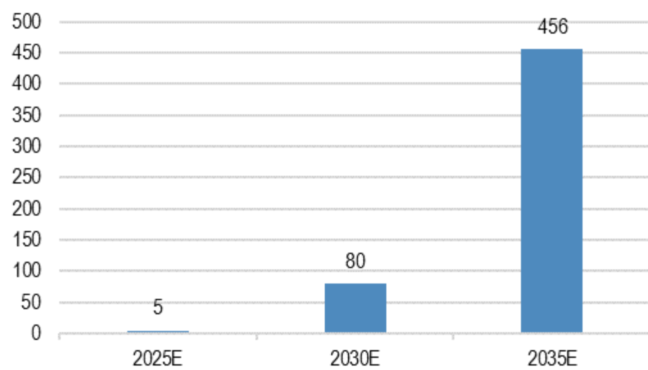
Source: GGII estimates, J.P. Morgan. Note: 2024 and 2025-26E data based on GGII estimate of effective capacity. Note: solid-state battery demand includes both semi-SSB and ASSB.

Figure 2: Solid-state battery demand forecast by GGII (Gwh)



Source: GGII estimates, J.P. Morgan. Note: solid-state battery demand includes both semi-SSB and ASSB. GGII estimate ASSB demand at 5Gwh in 2035E

Figure 3: Solid-state battery demand forecast by ICCSino (Gwh)



Source: ICCSino estimates, J.P. Morgan. Note: solid-state battery demand includes both semi-SSB and ASSB.

ASSB equipment orders so far

Table 5: ASSB equipment order flow summary

Company	ASSB Orders	Key Clients & Notes
Hangke	FY25 forecast: 50mn–0.1 bn (pilot lines)	BYD, Geely, EVE, Gotion, LGES, etc. Main clients in ASSB are domestic
Hymson	FY24: 0.2–0.3 bn; No stats for 2025	CALB, EVE, Cornex; Samsung, LGES, Toyota; Main clients in ASSB are domestic
Lyric	FY25 forecast: Several mn~0.1bn	GAC Aion, eVTOL; Main clients in ASSB are domestic
Leading player	1H25: 0.4-0.5bn; (FY24: 0.1bn); 2H25 higher than 1H25	Overseas clients are more aggressive but also seeing domestic client orders from 2H25.

Source: Company data.

Current ASSB orders

- Orders in the SSB business cover both semi-solid and all-solid battery technologies, with oxide and sulfide electrolyte types being explored. Companies identified semi-solid as still the main part in their SSB orders.
- ASSB orders are growing but still small. All three players report increasing ASSB-related orders but predict new order in FY25 lower than 0.1bn. ASSB still represents a small portion of total business. **Most ASSB orders are for prototypes, pilot lines, or small-scale production, not yet for full mass production.** Clients are actively testing SSB equipment, with some initial production lines and pilot lines delivered.
- ASSB equipment is more complex and valuable than traditional liquid battery equipment**. Industry players point out that ASSB equipment value is 2–5x higher than liquid ones in the current phase.**
- **Application:** SSB technology is being tested for a range of applications: energy storage, consumer electronics, power batteries, and even robotics. Some players expect SSB projects to gradually support future orders from their existing large clients and new overseas clients, and also as new application scenarios emerge (e.g., eVTOL, smart warehousing, robots).

Mass production of ASSB still 2–3 years away

- ASSB orders are expected to **increase significantly** after 2–3 years, as technology matures and clients finalize their technical routes.
- For now, SSB remains a **strategic R&D and pilot business** for most equipment makers. Battery equipment companies also point out that downstream clients are rigorous in testing in the ASSB business and this takes a longer time for the value chain to move to the mass-production step.
- However, once technical hurdles are overcome and mass production begins, ASSB equipment demand could grow rapidly, given the higher value and complexity of the equipment. Companies are positioning themselves to capture this future growth by building relationships and supporting client trials now.
- **Downstream demand:** Most players are seeing the primary demand for all-solid-state battery (ASSB) equipment coming from domestic clients. While many companies are actively engaging with overseas customers, domestic orders remain the main growth engine for ASSB business at this stage.

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Table 6: Changes in battery cell manufacturing process and equipment

		Front end Electrode & Electrolyte production							Mid-end Cell assembly						Back end Cell finishing
Manufacturing process		Slurry mixing 搅拌机	Wet Coating and Drying 涂布机	Dry Mixing 强力混合机	Dry Coating 干法涂布设备	Electrolyte Coating 电解质涂布设备	Calendaring 辊压机	Electrolyte Laminating 热复合设备	Sliting &Die Cuting 分条横切机	Resin Frame Printing 胶框印刷机	Stacking w/o separator 无隔膜碟片机	Contacting & packaging 极耳焊接&包装	Electrolyte filling 注液机	Liquid Electrolyte gelation 烘烤设备	High pressure Formation 大压力化成分容机
Liquid eletrolyte battery		v	v				v		v			v	v		
Semi-Solid State	Quasi-liquid electrolyte 固液混合工艺	v	v			v	v		v			v	v		
	Gel-like electrolyte 原位固化工艺	v	v				v		v			v	v	v	
All-Solid-State	Sulfide-based 硫化物	F	F	v	v	v	v	v	v	v	v	v			v
	Oxide-based 氧化物	F	F	v	v	v	v	v	v	v	v	v			v
	Polymer-based 聚合物	F	F	v	v	v	v	v	v		v	v			v

Source: Company data, J.P. Morgan. Note: F=Feasible

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Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.
Initiated coverage Jun 25, 2025. All share prices are as of market close on the previous business day.

Date	Rating	Price (HK$)	Price Target (HK$)
26-Jun-25	OW	319.00	400
31-Jul-25	N	426.60	415
14-Sep-25	OW	432.80	530
05-Oct-25	N	610.50	600
16-Nov-25	N	542.00	575
30-Nov-25	OW	472.00	650

Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.
Initiated coverage Sep 27, 2018. All share prices are as of market close on the previous business day.

Date	Rating	Price (Rmb)	Price Target (Rmb)
12-Feb-23	OW	445.95	600
11-May-23	OW	232.52	300
11-Aug-23	OW	246.50	315
14-Sep-23	N	216.60	235
20-Oct-23	N	182.70	218
07-Feb-24	N	158.33	170
09-Feb-24	N	162.84	185
17-Mar-24	OW	181.00	220
17-May-24	OW	199.67	225
07-Oct-24	OW	251.89	300
23-Jan-25	OW	258.00	400
08-Apr-25	OW	215.22	320
15-Apr-25	OW	224.00	330
22-Apr-25	NR	231.36	--
26-Jun-25	OW	254.90	370
31-Jul-25	OW	277.09	380
14-Sep-25	OW	325.00	480
30-Nov-25	OW	373.20	520

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